Scientific Audit, Publication Closure Decision, and Final Work Order

Hole-Aware XXZ Squeezing (HAXS)

Stage 5C.2G-R3.2A.5 Complete Two-Host G0 Return

Prepared for supervisory review

1 September 2026 — Version 1.0

HEADLINE: AMBER — CLOSE THE ENGINEERING LOOP, THEN SUBMIT OR STOP WITHIN 28 DAYS

G0 verdict: GREEN. The current A5 two-host G0 return is accepted and a strict G1-only receipt is included.

As-is physics-paper verdict: RED. The present evidence is not defensible as a mobile-hole mechanism paper.

Publication-closure verdict: AMBER. One final exact–surrogate validity/failure-boundary study can produce a legitimate specialist paper; no further open-ended stage proliferation is approved.

Audit scope	Archive integrity, two-host G0 semantics, candidate and control-plane readiness, bounded test execution, mature HAXS scientific evidence, model validity, novelty, venue fit, and a hard publication-closure plan.
Primary current evidence	A5 complete G0 return; embedded A5 protocol; Host-A/Host-B JUnit and command records; embedded accepted Stage 5C.2F results; prior official G1 failure and deterministic-quadrature repair history.
Production science run by auditor	None. The official G1 and all G2–G4 scientific campaigns were not executed.
Resource assumption	Laptop/team-machine execution, 16–64 GB RAM, optional 6–16 GB GPU, resumable CPU/GPU campaigns, no assumed paid HPC.

Abstract

The Stage 5C.2G-R3.2A.5 package closes the immediate authorization-engineering problem. The outer return and nested protocol hashes reproduce; the candidate self-hash reconstructs; a pristine exact-root check passes; the two-host comparison recomputes from primary records with no identity mismatch or forbidden state; both supplied hosts contain 402 full and 169 targeted passing tests; 16 current-stage lifecycle tests and the complete 169-test targeted ledger reproduce in the audit environment; and an exact-key G1-only receipt is accepted by the current finalizer in an isolated control root without mutating the protocol. The package is therefore accepted for one deterministic G1 attempt only.

The publication conclusion is less favourable. The mature scientific result is a negative static-only minus mobile-plus-spin-density contrast for one selected $3 \times 3 \times 3$ stochastic surrogate: -0.4118 dB in the primary block and -0.4744 dB in the frozen confirmation block, with hierarchical intervals below zero. That is real within the saved surrogate outputs. It does not establish microscopic mobile-hole dynamics, causal component attribution, broad dimensional generalisation, constructive recovery, or a no-go result. The current moving-hole model changes an occupancy mask rather than coherently transporting a spin-carrying particle. Existing work already establishes that vacancies, disorder, spin-density coupling, and itinerancy materially affect squeezing, so a generic "holes degrade squeezing" paper would have weak novelty.

The brutally honest decision is: do not submit the present package as a physics paper. Run the included deterministic G1 once, then time-box one final 28-day scientific closure campaign built around transport-only calibration and untouched constrained exact/Krylov validation. A decisive validated domain or predictive failure boundary can support a submission to Physical Review A or Physical Review Research; a rigorous negative or methods result is better aligned with APS Open Science or Journal of Physics B. If no stable validity domain or nontrivial failure boundary appears, stop the physics line and pursue only a cleaned software/metapaper or archive the project.

Contents

1 Executive Verdict and the Direct Publication Answer

1.1 Three separate verdicts

Decision layer	Verdict	Supervisory decision
Current A5 G0 evidence	**GREEN**	Accept the current two-host G0 return. The candidate is eligible for the supplied strict G1-only receipt.
Current as-is physics manuscript	**RED**	Do not submit a mechanism, recovery, no-go, or lithium mobile-hole paper from the present evidence. It would be scientifically vulnerable and likely rejected.
Next iteration	**AMBER**	Run one G1 closure attempt, then one final exact–surrogate publication campaign under a 28-day hard deadline.
Long-term high-impact potential	**PURPLE**	Possible only if microscopic validity, fixed-count controls, held-out prediction, and explanatory theory succeed. This is not the immediate target.

Brutally honest answer. The project has not produced a defensible physics paper yet. Much of the elapsed effort improved software governance rather than the scientific claim. The mature numerical signal is too narrow and the moving-hole model is not yet microscopically validated. Submitting the current result as a mobile-hole mechanism paper would risk a justified rejection and could damage confidence in the project.

What can still be published. A focused paper on the *domain of validity and failure boundary of stochastic mobile-hole surrogates for spin squeezing* is realistic if one final untouched exact/Krylov study is decisive. This route uses both positive and negative outcomes productively. It requires days to weeks, not another sequence of open-ended protocol versions.

1.2 Immediate supervisor actions

1. Accept Stage 5C.2G-R3.2A.5 G0.
2. Use the supplied `SUPERVISOR_AUTHORIZATION_G1_ONLY_STAGE5C2GR32A5.json`.
3. Run official deterministic G1 exactly once and stop.
4. Rename the scientific closure stage **HAXS-PC1: Exact–Surrogate Publication Closure**. Do not create R3.2A.6, A.7, or another authorization substage unless G1 exposes a genuine deterministic defect.
5. Complete transport calibration and untouched exact–surrogate validation, decide the paper route on Day 17, and submit or stop by Day 28.

1.3 The hard closure rule

No more indefinite iteration. The next 28 days end in exactly one of three outcomes:

1. **Validated-domain paper:** a frozen surrogate passes untouched microscopic validity in a declared domain.
2. **Failure-boundary paper:** the surrogate fails systematically, but the failure is predictable and outperforms trivial hole-count/connectivity baselines.
3. **Stop/pivot:** neither a valid domain nor a stable nontrivial boundary exists; submit only a cleaned software/metapaper if reuse criteria are met, otherwise archive the project.

There is no fourth outcome called “run another protocol-repair stage”.

2 Materials, Integrity, and Provenance

2.1 Material inspected

The audit inspected the current complete G0 return, its checksum sidecar, the local recovery record, the nested A5 protocol archive, the candidate, root and runtime manifests, immutable wheel and environment records, G1 configuration and registries, Host-A and Host-B JUnit and command records, the two-host comparator, finalizer, isolated launcher, lifecycle tests, adversarial outcomes, embedded Stage 5C.2F result archive, Stage 5C.3-VB design, and the previous A4B Phase-A report.

2.2 Archive and identity results

Object	SHA-256 / result	Status
Complete G0 return ZIP	`d9595f46dc919f8d3906b4909cd18833a3659e5f15a04f48ee39787c0f11c9f6`	Verified
Nested A5 protocol ZIP	`d804a203567419f4775fb1d7357cfd9675563ba31068bc10918b9d27f2e1f70f`	Recomputed
Candidate	`481ca1905bedc68d6ac3eb36ac61b80356d4e32cf8847b3e927f081201240932`	Self-hash reproduced
Logical complete-return record	`e4c81afb8eb00c60f24572e8e985e5077fda1445a977c05a555fed129ea7d91a`	Recomputed
Two-host comparison	`9745c61a7bbb358472f6d64832af48823686cad1af712e6ddefb3965b6008154`	PASS, recomputed
Pristine immutable root	775 content files, 185 directories, bytecode-closed	PASS

The outer archive contains 29 files and extracts without path traversal, duplicate-name, symlink, or encryption hazards. The nested protocol contains 776 files and 185 directories. Approximately 130 paths carry explicit historical or predecessor-stage signals. This is acceptable for an evidence bundle but unsuitable for a public software repository.

2.3 Host evidence and recovery provenance

Both host records bind the same candidate, protocol, runtime tree, wheel, environment, configuration, plan, unit registry, runner, and named-test ledger. The physical-host commitments differ. The comparison has no identity mismatch and no forbidden state.

The student states that the GitHub Host-B G0 run itself passed, but its post-G0 packager initially resolved the checkout root rather than the immutable protocol root. The final return was recovered locally from the immutable Actions artifact using the candidate-bound packager. The disclosure is appropriate. The present upload contains the recovered content-addressed return and a short recovery record; it does not separately contain the original downloadable Actions artifact or complete hosted log archive. This is a provenance caveat, not a reason to reject G0, because the finalizer recomputes the host semantics from the supplied primary records. The raw hosted artifact should nevertheless be archived with the eventual public release.

3 A5 Software, G0, and Receipt Audit

3.1 Two-host tested behaviour

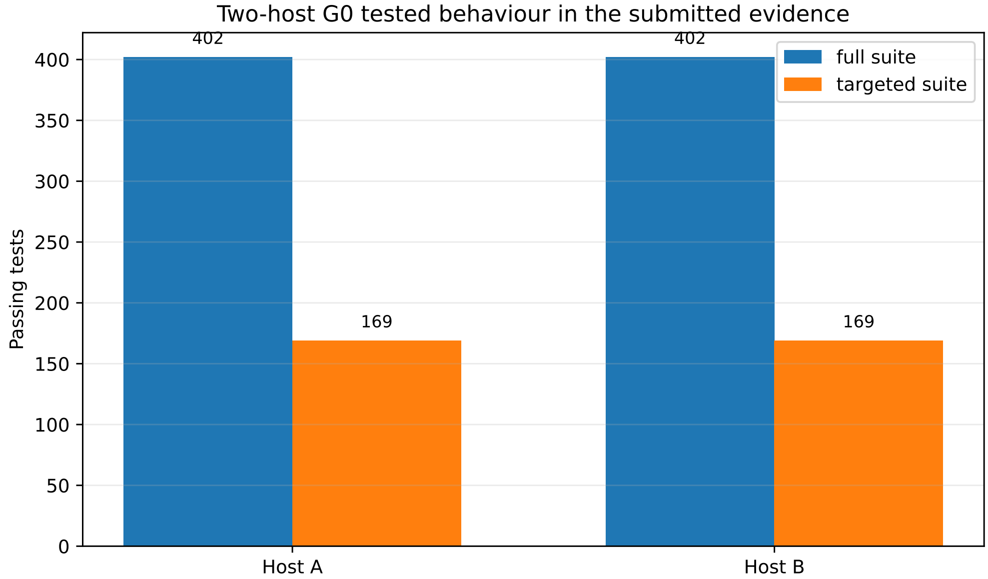

Figure 1: Supplied Host-A and Host-B JUnit records. Both hosts report 402 full and 169 targeted passing tests, with no failure, error, or skip. This supports tested protocol behaviour and environment reconstruction; it does not validate the mobile-hole physics.

The structured command records report zero exit status and expected/observed test-count agreement. Exact testcase identities and order agree with the candidate-bound ledgers. The current A5 package retains 60 declared fail-closed fixtures and four golden lifecycle fixtures.

3.2 Bounded independent execution

The audit reproduced:

- static Python compilation of the tracked source;
- 16/16 current A5 lifecycle tests;
- 169/169 candidate-bound targeted tests;
- candidate self-hash reconstruction;
- pristine exact-root verification;
- complete-return manifest and semantic recomputation;
- the two-host comparison from primary host records; and
- receipt validation and finalizer commitment in a temporary external control root.

The complete 402-test suite was not independently completed under the exact frozen Darwin ARM64 CPython 3.12.7 environment. The audit host used Linux CPython 3.13.5. The supplied JUnit XML and command records are therefore the primary evidence for the full-suite counts.

3.3 The A4 lifecycle contradiction is repaired

The previous A4 candidate wrote `AUTHORIZATION.json` into an immutable tree and then rejected that required file as an undeclared extra. A5 separates:

- an immutable protocol/data plane; and
- an external candidate-namespaced mutable control plane for receipt, authorization, setup, state, transcript, and artifacts.

The A5 finalizer validated the generated receipt and produced `LOCKED_G1_ONLY` in an audit-only control root. The immutable protocol root remained unchanged. The authorization records a single-attempt limit and requires setup/environment preflight before attempt reservation.

G0 decision: accepted. The current A5 package is sufficiently strong for one strict G1-only authorization. Continuing to search for hypothetical protocol defects after this point would create greater project risk than running the low-cost deterministic gate.

4 Scientific Evidence Actually Available

4.1 What A5 adds scientifically

A5 adds no new spin-squeezing result. Its records explicitly state that no scientific execution occurred and G1 was not authorized in the submitted package. A5 is an authorization and reproducibility milestone.

4.2 Mature selected-geometry result

The strongest accepted numerical result is embedded from Stage 5C.2F. For a selected $3 \times 3 \times 3$ stochastic surrogate, define

$$\Delta\xi^2_{\mathrm{dB}} = \xi^2_{\mathrm{dB}}(\text{static-only}) - \xi^2_{\mathrm{dB}}(\text{mobile+spin-density}).$$

Negative values indicate that the combined surrogate yields worse squeezing.

Table 1: Mature selected-geometry result from the repaired physical hierarchy. Intervals are the supplied hierarchical intervals.

Block	Physical occupancies	Mean (dB)	Hierarchical SE (dB)	95% interval (dB)
Primary	16	-0.4118	0.0487	[-0.5176,-0.3078]
Frozen confirmation	12	-0.4744	0.0394	[-0.5718,-0.3768]

All occupancy-level means are negative in both blocks. The primary–confirmation difference is approximately +0.0626 dB, and the supplied 90% equivalence interval lies within the declared ±0.25 dB margin.

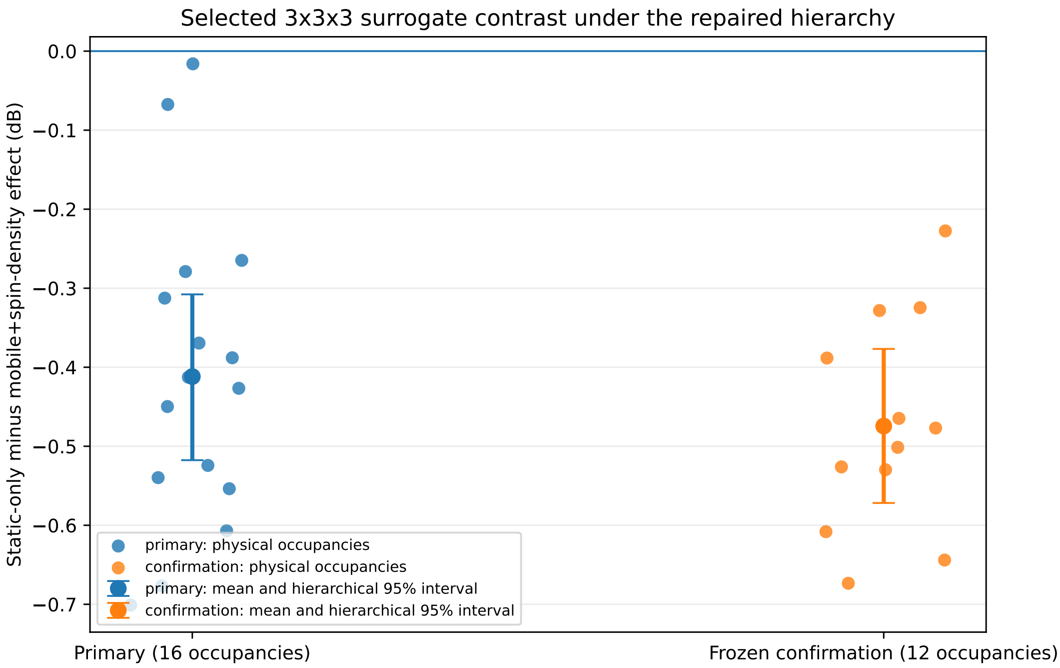

Figure 2: Physical-occupancy effects and block-level intervals for the selected $3 \times 3 \times 3$ surrogate. The negative direction replicates within this model and geometry. The figure does not establish microscopic validity, mechanism causality, or cross-geometry generalisation.

4.3 Uncertainty decomposition

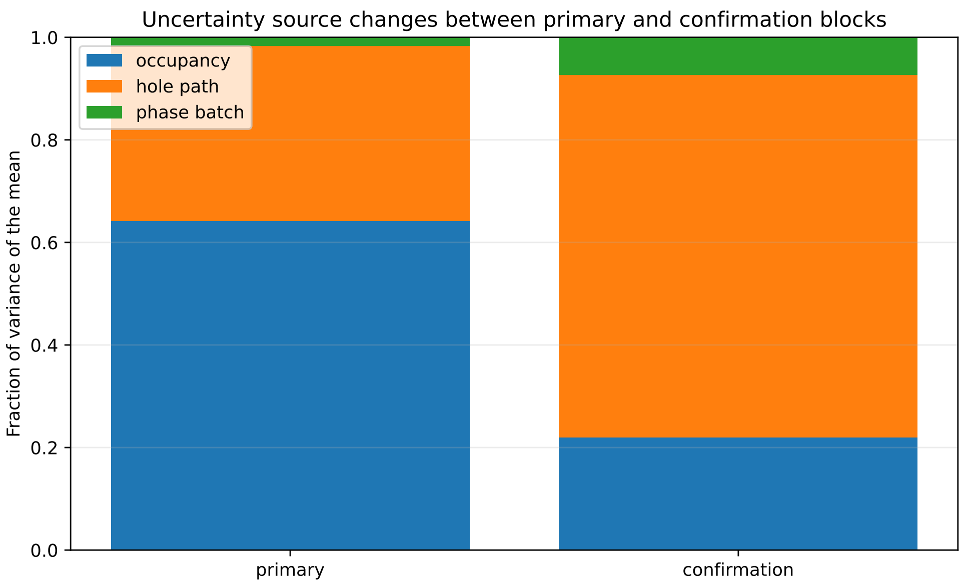

Figure 3: Fraction of the variance of the block mean attributed to occupancy, hole-path, and phase-batch levels. The dominant source changes between primary and confirmation, showing why raw trajectory count alone is not a universal precision remedy.

The primary block is occupancy-dominated; the confirmation block is path-dominated. This is a well-controlled Monte Carlo result for the chosen surrogate. It is not a validation of the surrogate's physical content.

4.4 Why the mechanism claim remains blocked

The current mobile-hole implementation evolves a stochastic occupancy mask. Spins remain site-indexed and are masked/unmasked as holes move. This is not equivalent to coherently transporting a spin-carrying particle under a constrained hopping Hamiltonian, and it does not reproduce carrier–spin entanglement by construction. The explicit spin-density field is a phenomenological function of nearby holes. The package also contains a constrained exact/Krylov model and transport observables, but the required transport-only calibration and untouched comparison have not yet been executed.

The literature makes this distinction consequential. The contact-interaction experiment already reports a pronounced 1D/3D difference and identifies finite holes, hole motion, and direct spin-density coupling as important ingredients [1]. Hole doping and mixed-dimensional strategies have been analysed theoretically [2, 3]; recent disorder work maps squeezing thresholds [4]; and itinerancy has been observed to improve squeezing in a different dipolar t-J regime [5]. Mobility is therefore not generically harmful. A publishable HAXS result must identify *when and why* the surrogate agrees or fails, not merely report a negative contrast.

5 Claim–Evidence Audit

Table 2: Current claim controls. The full machine-readable ledger is supplied as `claim_audit.csv`.

Claim	Verdict	Strongest allowable wording	Forbidden wording / required repair
A5 two-host G0 passed	Supported	The content-addressed A5 host records reconstruct the same frozen execution object and pass the recomputed comparator.	Do not call this scientific replication.
402 full and 169 targeted tests passed on each host	Supported from primary records	Supplied exact JUnit records contain 402 and 169 passing tests with no failure, error, or skip.	Do not say the full suite was locally reproduced on Linux/CPython 3.13.
Candidate is ready for a G1-only receipt	Supported	The current exact-key receipt validates and commits in an isolated control root.	G2 and later scopes remain blocked.
A negative $3 \times 3 \times 3$ surrogate effect exists	Supported narrowly	Primary and confirmation hierarchical intervals lie below zero for the selected surrogate.	Do not generalise across dimensions or attribute causally to microscopic hole motion.
Mobile holes cause the degradation	Unsupported	The data are consistent with degradation in a moving-occupancy surrogate.	Requires transport-only calibration and untouched exact/Krylov validation.
Constructive recovery was achieved	Contradicted	The constructive route was not achieved under the tested programme.	No recovery or 3 dB claim.
The present package is a publishable physics paper	Unsupported	It is a strong internal computational artifact with a narrow surrogate result.	Complete M01/M02 or use a software/methods route.

Claim	Verdict	Strongest allowable wording	Forbidden wording / required repair
The code is ready for a software journal	Unsupported now	The code has extensive tests and provenance machinery.	Add an OSI licence, clean user repository, documentation, worked example, archive, and public-development evidence.

6 Brutal Publication-Readiness Audit

6.1 Would I submit the current physics paper now?

No. A manuscript whose central claim is that mobile holes and spin-density coupling degrade 3D squeezing would face five immediate objections:

1. the mobile-hole model is not a microscopic carrier-spin model;
2. the strongest result is one repeatedly selected geometry;
3. the effect is modest and not calibrated to the experimental discrepancy;
4. fixed-hole/topology and untouched microscopic controls are incomplete;
5. the broad qualitative premise overlaps existing experimental and theoretical work.

PRL requires a pivotal advance, an essential step on a critical problem, a highly consequential method, or unusual broad interest [6]. PRX Quantum requires an exceptional advance, connection, capability, or insight [7]. The current work meets neither standard.

PRA and Physical Review Research require a significant, authoritative, substantive addition [8, 9]. The current one-shape, unvalidated surrogate result is also below that threshold.

6.2 Can something be published?

Yes, but not by relabelling the current result. Two legitimate routes remain.

Recommended physics route. *Domain of validity of stochastic mobile-hole surrogates for spin squeezing in doped XXZ magnets.* Calibrate motion using transport only, freeze the mapping, and test untouched constrained exact/Krylov cases. A positive result yields a restricted validated surrogate. A negative result yields a quantitative failure boundary. Both are useful if the outcome is stable and beats trivial baselines.

Software/methods fallback. *HAXS: a fail-closed validation and hierarchical uncertainty framework for stochastic spin-squeezing simulations.* This route requires a clean public repository, an OSI-approved licence, a stable API/CLI, one nontrivial worked example, documentation, a DOI, and evidence of reuse potential. The current package lacks those release essentials. JORS specifically seeks reusable research software with high reuse potential [10]. SciPost Physics Codebases expects executable code, tests, intelligible documentation, and a nontrivial reproducible example. JOSS is not an immediate route because its current screening includes public development history and demonstrated impact signals, and the repository lacks a licence and public-use record [11, 12].

6.3 Venue ladder

Venue route	Current decision	Conditions
PRL / PRX Quantum	Reject now	Only reconsider after broad held-out prediction, theory, and independent or experimental validation.
Physical Review A	Conditional first target	Use if M02 identifies a substantive validated domain or sharp mechanism/failure boundary with adequate breadth.
Physical Review Research	Conditional first/second target	Use if the contribution is methodological and broadly relevant across physics, with strong exact validation.
APS Open Science	Strong fallback	Explicitly welcomes software advances, methods, replication, and negative/null results, but still requires valid, reliable and useful science [13].
Journal of Physics B	Strong specialist fallback	Welcomes theoretical and numerical cold-matter/AMO work; subscription publication can avoid an OA fee [14].
JORS / SciPost Physics Codebases / SoftwareX	Software fallback	Only after S02 release cleanup and a reusable user-facing package.
JOSS	Not immediate	Current public-history, impact, licence and open-development requirements are not met.

7 Negative-Result Autopsy and Rescue Hypotheses

Table 3: Negative-result autopsy.

Observed result	Plausible diagnosis	Discriminating test	Scientific value
No constructive recovery	The tested control class genuinely lacks a robust recovery; or search/parameter mapping was incomplete.	Do not reopen this route during closure. State the tested scope and move to validity/failure-boundary analysis.	A clean negative prevents further sunk cost.
Negative surrogate contrast	Real mobility/spin-density obstruction; occupancy-mask artefact; or hole-count/topology confounding.	M01/M02 exact validation and M03 matched fixed-count/topology controls.	Positive mechanism, predictive failure boundary, or falsification of the surrogate.
One selected geometry dominates	Selection bias and finite-size dependence.	Freeze and test untouched exact cases first; broader geometry only if M02 passes.	Restricts claims honestly and prevents a false universal story.
Long sequence of software repairs	Initial code and inference were under-specified; later controls improved reliability but displaced physics.	Stop creating authorization substages after G1.	Converts the accumulated software work into a clean artifact rather than continuing the loop.

Observed result	Plausible diagnosis	Discriminating test	Scientific value
Software-paper readiness weak	Internal evidence bundle is not a user package.	S02 external clean install and worked example.	Provides a lower-risk publication route if the physics fails.

8 The Final 28-Day Publication-Closure Plan

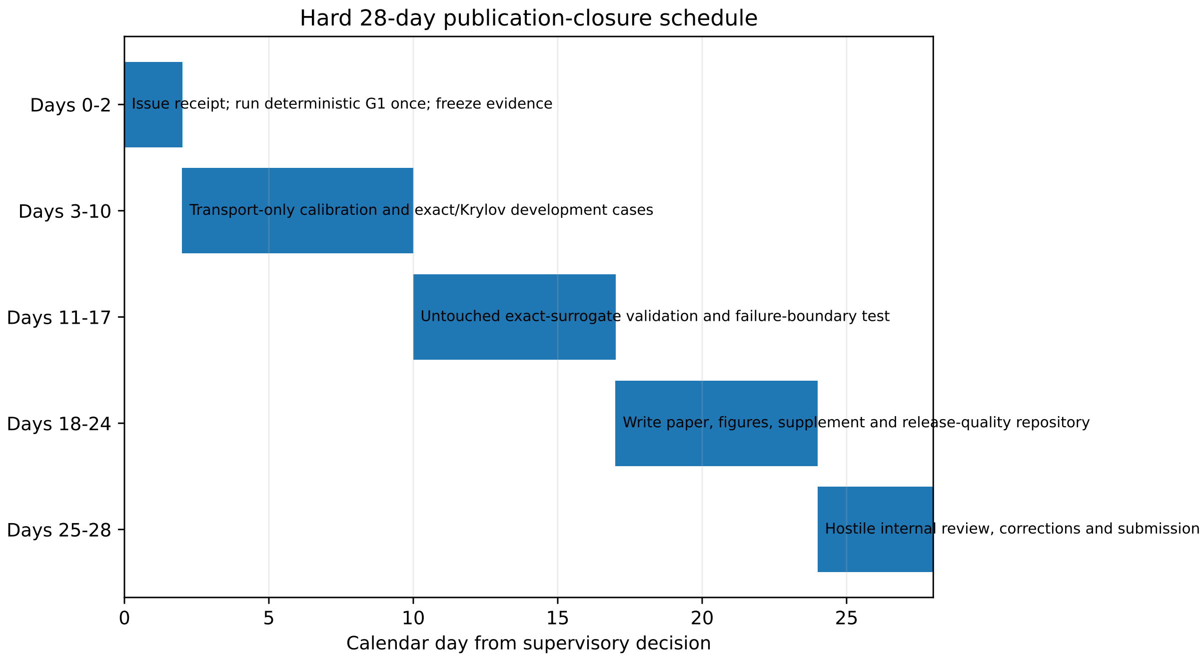

Figure 4: Hard closure schedule. The decision point occurs on Day 17. The plan ends with submission or explicit stop; it does not roll into another indefinite stage chain.

8.1 Days 0–2: close G1

Use the included receipt. Run the deterministic G1 once. Preserve all evidence. If G1 passes, close protocol engineering. If G1 fails, allow one 48-hour root-cause review only. A genuine deterministic scientific defect blocks M01. A packaging-only failure is repaired once without changing the scientific estimator; then the project proceeds or stops.

8.2 Days 3–10: transport-only calibration

Run M01 on small exact systems. Squeezing is forbidden during calibration and model selection. Freeze the mobility map, transport tolerances, untouched registry, and analysis code before opening M02.

8.3 Days 11–17: untouched rescue-or-kill test

Run M02. The outcome must answer:

1. Does the surrogate reproduce transport?
2. Does it preserve the sign of the squeezing contrast?
3. Does it preserve mobile-only versus spin-density-only ranking?
4. Does it reproduce the time-profile shape within a declared tolerance?
5. Is any valid/failure region predictable beyond hole-count and connectivity baselines?

8.4 Day 17 decision

Outcome	Manuscript route
All central untouched criteria pass in a nontrivial domain	Validated-domain paper; submit first to PRA or PRR.
Untouched validity fails, but failures form a stable held-out boundary	Failure-boundary paper; submit to PRA/PRR if broad, otherwise APS Open Science or Journal of Physics B.
No stable valid domain or nontrivial boundary; trivial baselines explain the result	Stop the physics claim. Complete S02 and submit only a software/metapaper if reuse criteria are met; otherwise archive.

8.5 Days 18–28: write and submit

The manuscript must be drafted from a frozen evidence manifest. No additional exploratory campaign is permitted unless it repairs a clearly identified fatal omission and can complete within the 28-day deadline.

9 Small, Medium, and Big Experiments for the Next Iteration

The machine-readable `experiment_ladder.json` contains complete experiment cards. The closure-critical experiments are S01, M01, and M02. The remaining cards are contingency or future-impact routes, not permission to extend the current project indefinitely.

Table 4: Experiment ladder.

ID	Size / tier	Priority	Objective	Pass criterion	Claim unlocked
S01	Small / 1	P0	One official deterministic G1	All equality and deterministic-sanity predicates pass; semantics agree	Implementation consistency only
S02	Small / 1	P0	Clean public software release	Two independent clean installs reproduce a worked example	Software-paper eligibility
S03	Small / 0	P0	One manuscript evidence manifest	Every number and figure regenerates without manual editing	Reproducible drafting
M01	Medium / 2	P0	Transport-only mobility calibration	Frozen map meets transport tolerances and window sensitivity	M02 opening
M02	Medium / 2	P0	Untouched exact–surrogate validity	Sign, ranking, time profile and transport pass, or failures form a stable predictive boundary	Physics closure paper

ID	Size / tier	Priority	Objective	Pass criterion	Claim unlocked
M03	Medium / 2	P1	Fixed-count/topology controls	Effect persists after matched count/topology adjustment	Stronger large-system case
B01	Big / 4	P1	Validity/failure phase map	Held-out boundary beats trivial baselines	Broad PRA/PRR route
B02	Big / 4	P2	Unseen dimension/geometry prediction	Frozen predictor generalises across 1D/2D/3D	Possible high-impact route
B03	Big / 5	P2	Prospective experimental prediction	Blinded experiment matches preregistered prediction	Strong high-impact evidence

10 Manuscript Blueprint

10.1 Recommended paper title and claim

Recommended title:
Domain of validity of stochastic mobile-hole surrogates for spin squeezing in doped XXZ magnets
Core claim after M02:
A transport-calibrated moving-occupancy surrogate reproduces selected squeezing observables only within a quantified domain; outside that domain it fails predictably. The paper reports the domain, the failure modes, and the consequences for interpreting large-system semiclassical simulations.

This framing is honest whether the surrogate succeeds or fails. It also avoids claiming exact lithium dynamics unless the comparator supports that statement.

10.2 Main-figure plan

1. **Model hierarchy and inference units.** Static vacancies, stochastic moving occupancy, spin-density field, and constrained exact/Krylov comparator; show what is and is not represented.
2. **Transport-only calibration.** Exact and surrogate hole-density propagation, mean-square displacement, return probability, frozen mobility map, and uncertainty.
3. **Untouched validity.** Exact versus surrogate squeezing time profiles; sign and component-ranking matrix; deliberately misspecified controls.
4. **Domain or failure boundary.** Held-out classification and calibration versus hole-count/connectivity baselines.
5. **Large-system case study.** The accepted $3 \times 3 \times 3$ primary/confirmation contrast and uncertainty decomposition, explicitly labelled as a stochastic-surrogate application.

10.3 What should stay in the supplement

Protocol authorization, candidate hashes, adversarial receipt fixtures, superseded stage chronology, complete random-effects derivation, seed registries, full validation tables, environment records, and release manifests should be supplementary or artifact material. They must not dominate the scientific narrative.

11 Codebase and Community-Utility Decision

The codebase contains extensive tests, deterministic packaging, manifests, hierarchical uncertainty tools, exact and stochastic models, and fail-closed gates. Those are genuine assets. The current object is nevertheless not a public package:

- no OSI-approved licence file;
- no `CITATION.cff`;
- no contribution guide or changelog;
- a stale top-level README that still describes A4;
- many predecessor archives and stage-specific paths;
- no stable user-facing API/CLI contract;
- no independent user installation;
- no demonstrated external adoption.

The public repository should contain only the stable library, examples, tests, documentation, and release metadata. The full audit/custody bundle should be deposited separately as a citable artifact.

12 Risks and Hostile Reviewer Objections

Table 5: Highest-priority risks and responses.

Risk / objection	Why it matters	Required response	Stop trigger
Moving mask is not microscopic motion	Causal mechanism language is invalid without carrier-spin transport	M01/M02 exact/Krylov benchmark	Untouched sign/ranking/time profile fail without stable boundary
One geometry carries the result	Selection bias and no generalisation	Make it a case study; use untouched small-system registry	Claim still depends on selected geometry
Effect may be topology/hole-count driven	Mobility attribution is confounded	M03 or restrict claim to exact-surrogate validity	Simple graph baseline explains result
Protocol engineering overwhelms physics	Reviewers do not reward internal governance as a physics advance	Move details to supplement and artifact	Manuscript introduction leads with G0/G1 machinery
Literature overlap	"Holes degrade squeezing" is already anticipated	Domain/failure boundary and predictive mechanism	No novelty beyond generic degradation
Further stage drift	Opportunity cost now exceeds benefit	28-day hard closure	Any A6/A7 request without M02 evidence
Software release is not reusable	Software venues require open, documented, installable tools	S02 release cut	External user cannot reproduce example

13 Exact Next-Iteration Work Order

13.1 Stage name

HAXS-PC1 — Exact–Surrogate Publication Closure.

13.2 Mandatory sequence

1. Use the supplied G1-only receipt; run A5 G1 once; return terminal evidence.
2. Freeze the failed R3.1 and all superseded candidate histories as appendix artifacts.
3. Create a clean paper repository and evidence manifest.
4. Finalise M01 case registry, transport tolerances, untouched M02 registry, and analysis code before running M01.
5. Run M01; freeze the mobility map.
6. Run M02 exactly once on the untouched registry.
7. On Day 17 choose validated-domain, failure-boundary, or stop/software route.
8. Write, internally attack, correct, and submit by Day 28.

13.3 Forbidden work

- No new constructive-recovery campaign.
- No broad Stage 5D finite-size campaign.
- No new authorization stage unless G1 exposes a deterministic defect.
- No post-hoc squeezing fit in the transport calibration.
- No exact-mobile-hole, lithium-dynamics, universal no-go, or top-journal-readiness wording.
- No public software-paper claim before licence, documentation, release, and external installation.

13.4 Evidence the student must return

- G1 terminal control root and all raw artifacts.
- M01 and M02 frozen configurations and registries.
- Exact commands and environment/hardware records.
- Raw exact and surrogate curves, failed runs, and attempt ledger.
- Frozen mobility map and calibration uncertainty.
- Untouched sign/ranking/time-profile decisions.
- Baseline comparisons and negative controls.
- Source-data CSVs and generated figures.
- One manuscript evidence manifest and checksum ledger.
- A one-page interpretation that separates evidence, inference, and allowed wording.

14 Supervisor Action Matrix

Action	Decision	Basis
Accept A5 G0	**APPROVE**	Current identities, host semantics, JUnit evidence, comparator, root, and lifecycle checks support it.
Issue included G1-only receipt	**APPROVE**	Receipt validates in the current finalizer; every downstream scope remains blocked.
Run G1 once	**APPROVE**	Low-cost deterministic closure of implementation consistency.
Submit current mobile-hole physics paper	**REJECT**	No microscopic validity or broad generalisation.
Start HAXS-PC1 M01/M02	**APPROVE CONDITION-ALLY**	G1 must pass; protocol is frozen before scientific rows.
Run broad Stage 5D	**REJECT NOW**	It does not answer the central validity question.
Prepare PRA/PRR manuscript	**CONDITIONAL**	Only after a decisive M02 valid domain or nontrivial failure boundary.
Prepare APS Open Science/J. Phys. B fallback	**APPROVE**	Appropriate for a rigorous negative/methods result after M02.
Prepare software paper	**CONDITIONAL**	Requires S02 release readiness; JOSS is not immediate.
Continue protocol staging beyond Day 28	**STOP**	Opportunity cost is no longer justified.

15 Final Recommendation

Final recommendation: proceed once, then close.
The current A5 G0 return is accepted and the attached G1-only receipt is issued. Run G1 exactly once. Do not continue auditing the authorization machinery after a pass.
Do not submit the current data as a mobile-hole mechanism paper. The strongest honest scientific statement is a replicated negative contrast inside one selected stochastic moving-occupancy surrogate. That is not enough for a defensible physics publication.
Use one final 28-day campaign to determine the surrogate's microscopic domain of validity or its failure boundary. Submit a strong outcome to PRA or PRR. Submit a rigorous narrower or negative outcome to APS Open Science or Journal of Physics B. If M02 yields neither a valid domain nor a stable nontrivial boundary, stop the physics line and publish only a cleaned software/metapaper if the release criteria are actually met.
This is the last approved HAXS research iteration.

A Reproducibility Commands and Outcomes

```
# Current-stage lifecycle suite
python -m pytest -q tests/stage5c2gR32A5
# Result: 16 passed

# Candidate-bound targeted ledger
python -m pytest -q -p no:cacheprovider <169 frozen nodeids>
# Result: 169 passed

# Audit-only receipt finalization
python -I scripts/finalize_stage5c2gR32A5_authorization.py \
  --receipt SUPERVISOR_AUTHORIZATION_G1_ONLY_STAGE5C2GR32A5.json \
  --protocol-archive HAXS_Stage5C2G_R3_2A_5_Protocol.zip \
  --g0-return HAXS_Stage5C2G_R3_2A_5_Complete_G0_Return.zip \
  --control-root <TEMP_CONTROL_ROOT> --commit
# Result: LOCKED_G1_ONLY; protocol unchanged
```

The audit environment was Linux x86-64 with CPython 3.13.5. The reference environment is Darwin ARM64 with exact CPython 3.12.7. The full 402-test suite was therefore verified from primary host JUnit and command evidence rather than labelled as locally reproduced.

B Artifact Inventory Summary

Surface	Files	Approximate bytes
Complete G0 return	29	416,454,056
Nested pristine protocol	776	457,213,058

The complete per-file inventory, SHA-256 values, size, suffix, and stale/legacy hints are supplied in `full_file_inventory.csv` and `full_file_inventory.json`.

C Receipt Identity

Field	Value
Receipt file	`SUPERVISOR_AUTHORIZATION_G1_ONLY_STAGE5C2GR32A5.json`
Receipt ID	`f8035d48-a73c-49b1-bf60-872983b24d62`
Issued at UTC	`2026-09-01T16:41:42Z`
Receipt SHA-256	`16b1c098c6201a587e1929eae50999d5eb44fb923cecc41694a01a25562429db`
Candidate	`481ca1905bedc68d6ac3eb36ac61b80356d4e32cf8847b3e927f081201240932`
Authorized scope	`G1_ONLY`
Blocked scopes	G2, G3, G4, Stage 5C3, Stage 5D, manuscript-result claims, exact-mobile-hole claims, public release

D Machine-Readable Audit Assets

The master bundle contains:

- `evidence_ledger.csv/json`;
- `claim_audit.csv`;
- `negative_result_autopsy.csv`;
- `repair_matrix.csv/json`;

- `experiment_ladder.csv/json`;
- `publication_claim_ladder.csv`;
- `publication_route_matrix.csv`;
- `reviewer_objections.csv`;
- `risk_register.csv`;
- `prior_instruction_compliance.csv`;
- `full_file_inventory.csv/json`;
- `audit_summary.json`;
- figure source data and generation script; and
- exact commands, assumptions, compilation log, and page renders.

E Assumptions and Limitations

- The original hosted GitHub Actions artifact was not separately supplied in the current top-level upload; the audit used the recovered content-addressed Host-B evidence included in the complete return.
- The complete 402-test suite was not rerun under the exact reference platform.
- The official G1 and all G2–G4 scientific experiments were not executed by the auditor.
- The mature scientific analysis uses the embedded accepted Stage 5C.2F result archive and its source-generated tables.
- Venue positioning is current to 1 September 2026 and remains an editorial judgement, not an acceptance guarantee.
- Any AI-assisted analysis or manuscript preparation submitted to APS should follow the current APS disclosure policy and remain under human verification and accountability.

References

[1] Yoo Kyung Lee, Maxwell Block, Hanzhen Lin, Vitaly Fedoseev, Philip J. D. Crowley, Norman Y. Yao, and Wolfgang Ketterle. “Observation of Spin Squeezing with Contact Interactions in One- and Three-Dimensional Easy-Plane Magnets”. In: *Physical Review Letters* 135 (2025), p. 023402. DOI: 10.1103/jqvz-kpwg.

[2] Tanausú Hernández Yanes, Artur Niezgoda, and Emilia Witkowska. “Exploring spin squeezing in the Mott insulating regime: Role of anisotropy, inhomogeneity, and hole doping”. In: *Physical Review B* 109 (2024), p. 214310. DOI: 10.1103/PhysRevB.109.214310.

[3] Mikhail Mamaev, Diego Barberena, and Ana Maria Rey. “Spin squeezing in mixed-dimensional anisotropic lattice models”. In: *Physical Review A* 109 (2024), p. 023326. DOI: 10.1103/PhysRevA.109.023326.

[4] Samuel E. Begg, Bishal K. Ghosh, Chong Zu, Chuanwei Zhang, and Michael H. Kolodrubetz. *Scalable Spin Squeezing in Power-Law Interacting XXZ Models with Disorder*. 2026. DOI: 10.48550/arXiv.2601.10703. arXiv: 2601.10703 [quant-ph].

[5] Alec Douglas, Vassilios Kaxiras, Lin Su, Michal Szurek, Vikram Singh, Ognjen Marković, and Markus Greiner. “Spin Squeezing with Itinerant Magnetic Dipoles”. In: *Physical Review X* 15 (2025), p. 041021. DOI: 10.1103/shj7-9kb3.

[6] American Physical Society. *About Physical Review Letters*. 2026. URL: https://journals.aps.org/prl/about (visited on 09/01/2026).

[7] American Physical Society. *About PRX Quantum*. 2026. URL: https://journals.aps.org/prxquantum/about (visited on 09/01/2026).

[8] American Physical Society. *About Physical Review A*. 2026. URL: https://journals.aps.org/pra/about (visited on 09/01/2026).

[9] American Physical Society. *About Physical Review Research*. 2026. URL: https://journals.aps.org/prresearch/about (visited on 09/01/2026).

[10] Journal of Open Research Software. *Focus and Scope*. 2026. URL: https://openresearchsoftware.metajnl.com/about (visited on 09/01/2026).

[11] Journal of Open Source Software. *Review criteria*. 2026. URL: https://joss.readthedocs.io/en/latest/review_criteria.html (visited on 09/01/2026).

[12] Journal of Open Source Software. *Editorial guide and pre-review screening gates*. 2026. URL: https://joss.readthedocs.io/en/latest/editing.html (visited on 09/01/2026).

[13] American Physical Society. *About APS Open Science*. 2026. URL: https://journals.aps.org/apsos/about (visited on 09/01/2026).

[14] IOP Publishing. *About Journal of Physics B: Atomic, Molecular and Optical Physics*. 2026. URL: https://publishingsupport.iopscience.iop.org/journals/journal-of-physics-b-atomic-molecular-optical-physics/about-journal-physics-b-atomic-molecular-optical-physics/ (visited on 09/01/2026).